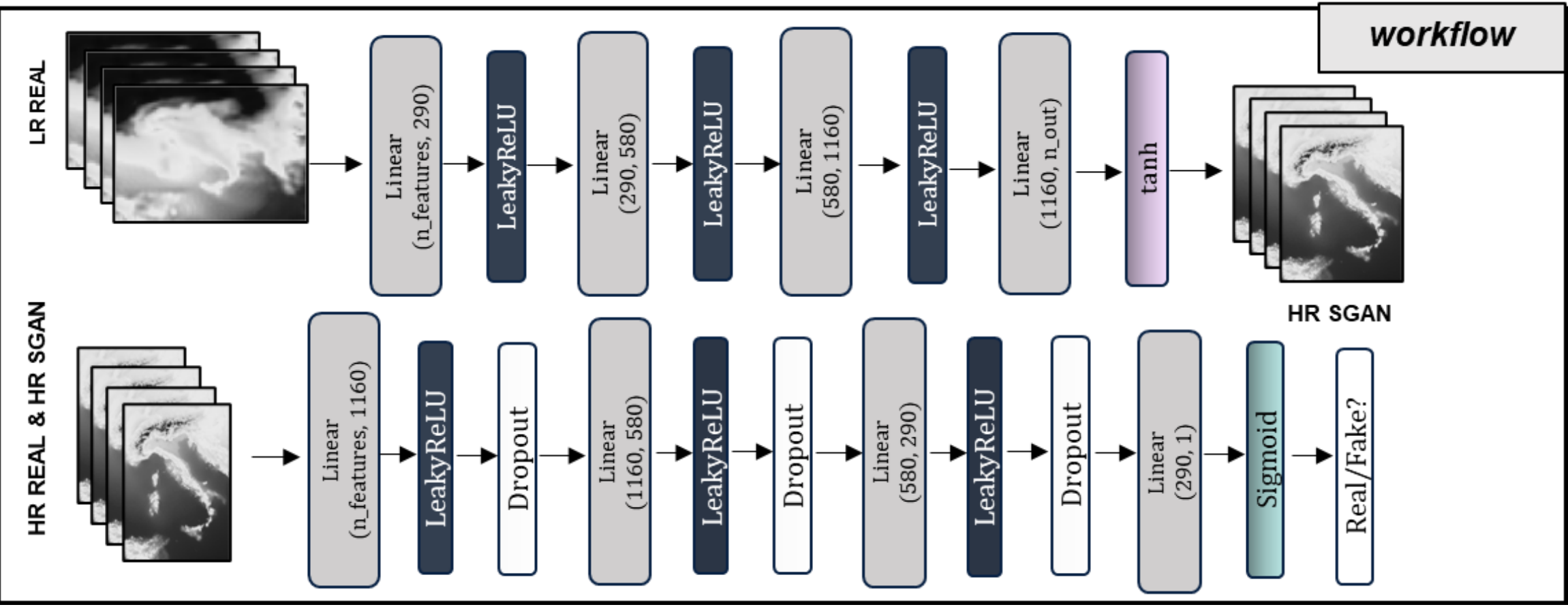


Outputs derived from downscaling techniques play a central role in developing a comprehensive understanding of localized impacts of climate variability and change, particularly in the context of extreme climate events. Local climate data are essential for improving risk assessments and formulating adaptation strategies across various sectors. This study presents a cGAN-based approach for downscaling of 2m-temperature and total precipitation from ~31 km to ~2.2 km resolution. The model shows high performance, capturing spatial patterns and extremes with reduced computational cost compared to dynamical models. Its flexibility and scalability make it a promising tool for generating high-resolution climate data across different regions.

A new Conditional GAN for climate downscaling

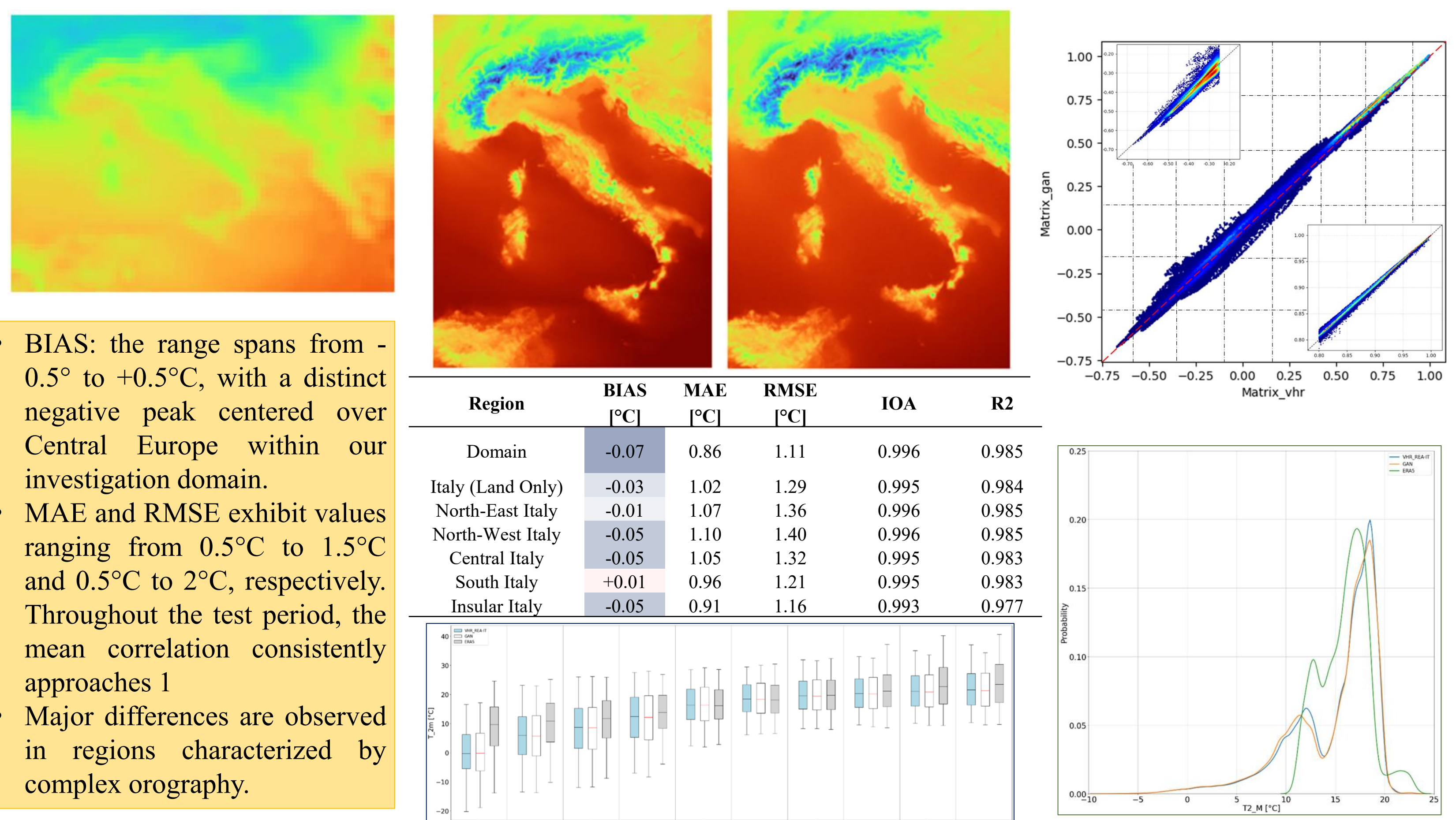


Traditional GANs, which generate data based solely on random noise, often struggle to incorporate external information, leading to less accurate or less relevant outputs for specific tasks. In contrast, cGANs can leverage high-resolution observational data or other auxiliary variables to produce more accurate and contextually relevant downscaled outputs.

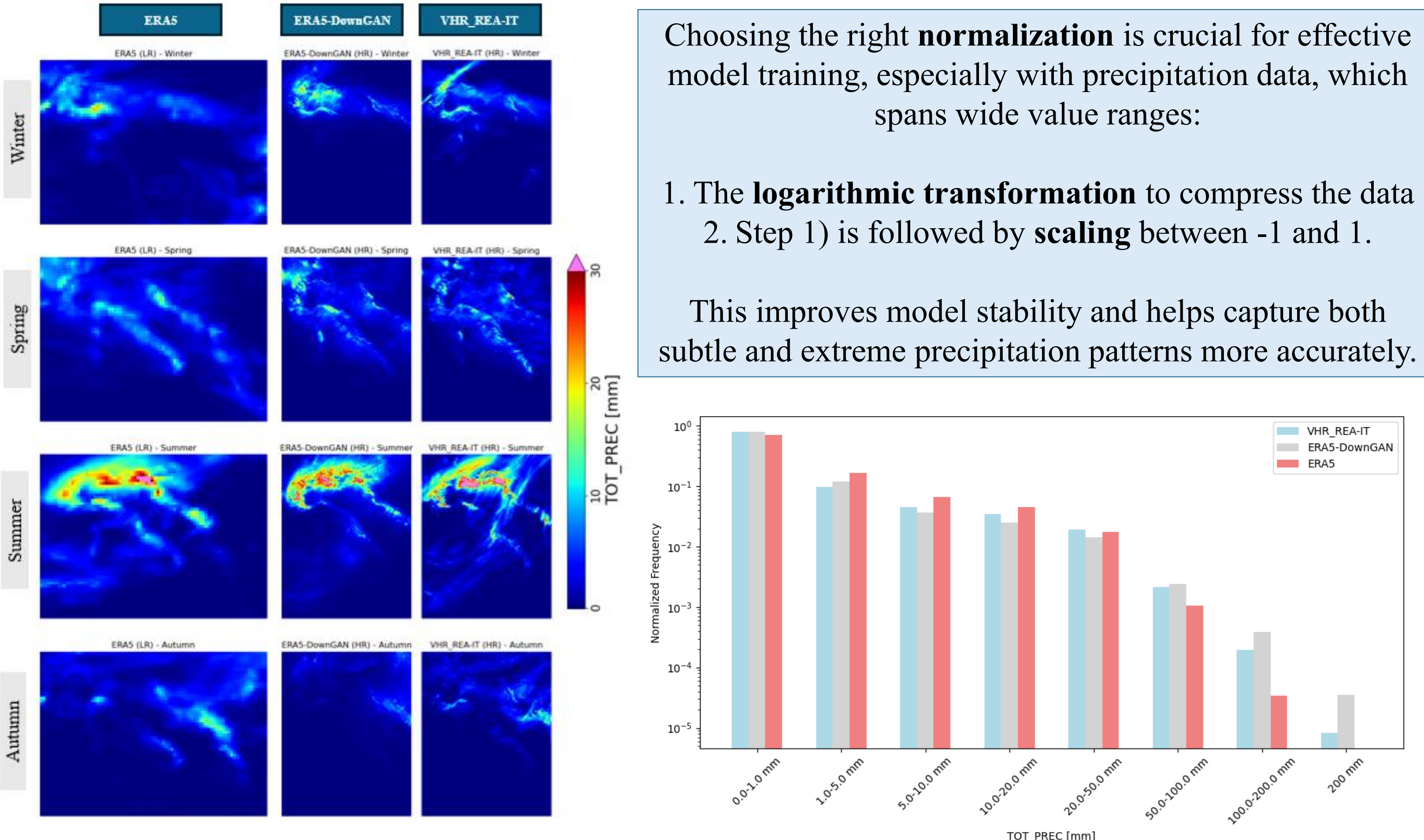
Training Dataset

- High resolution (HR)** : VHR-REA_IT dataset (Raffa et al., 2021), at 2.2 km of horizontal resolution.
- Low resolution (LR)** : ERA5 reanalysis data at a horizontal resolution of 0.25 degrees.

2m-temperature



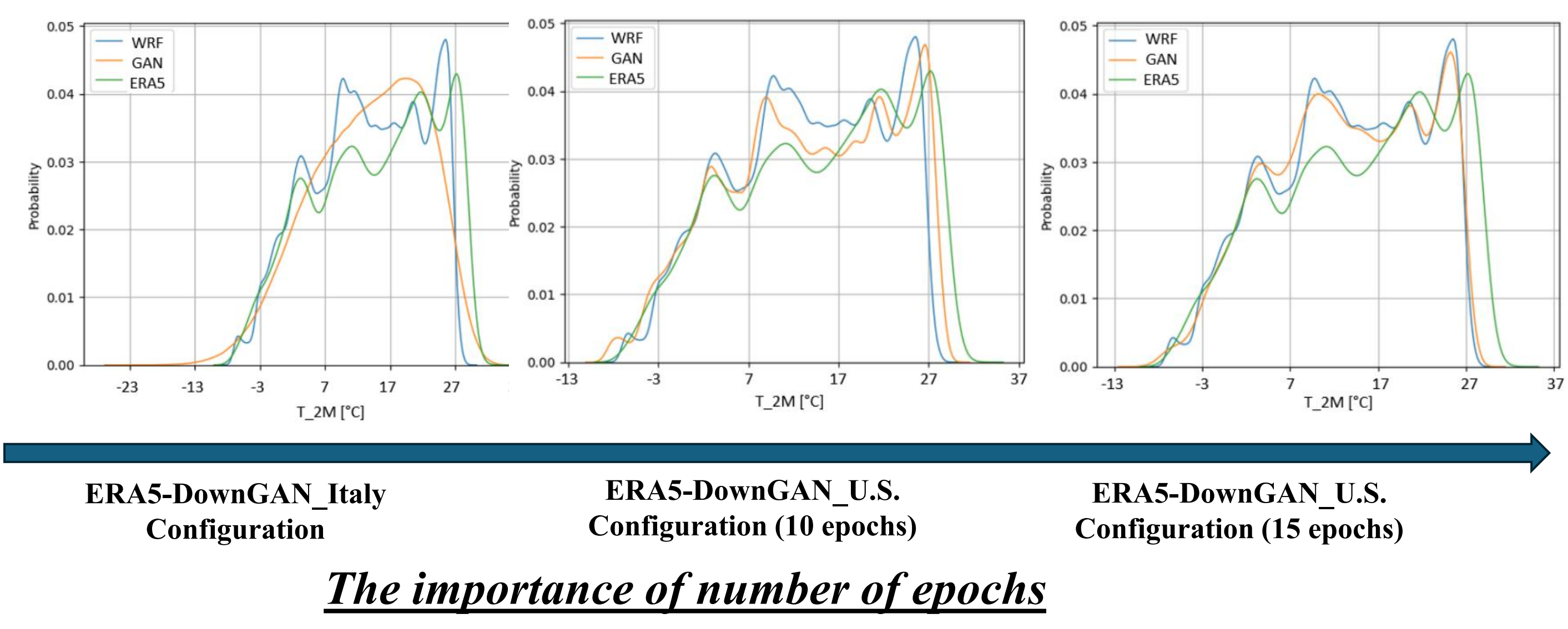
Total Cumulative Precipitation



U.S. Case Study

Extension of [Manco et al. \(2025\)](#) to the entire U.S. territory, in collaboration with the National Center for Atmospheric Research (NCAR). **Demonstrates:**

- Portability** of the **architecture** on different HPC systems (e.g., Casper)
- High performance across **large and diverse geographical domains**



It was essential to **increase the number of neurons** while maintaining the same architecture as the Italian case study, owing to the substantially larger dimensions of the geographical domain in the U.S.

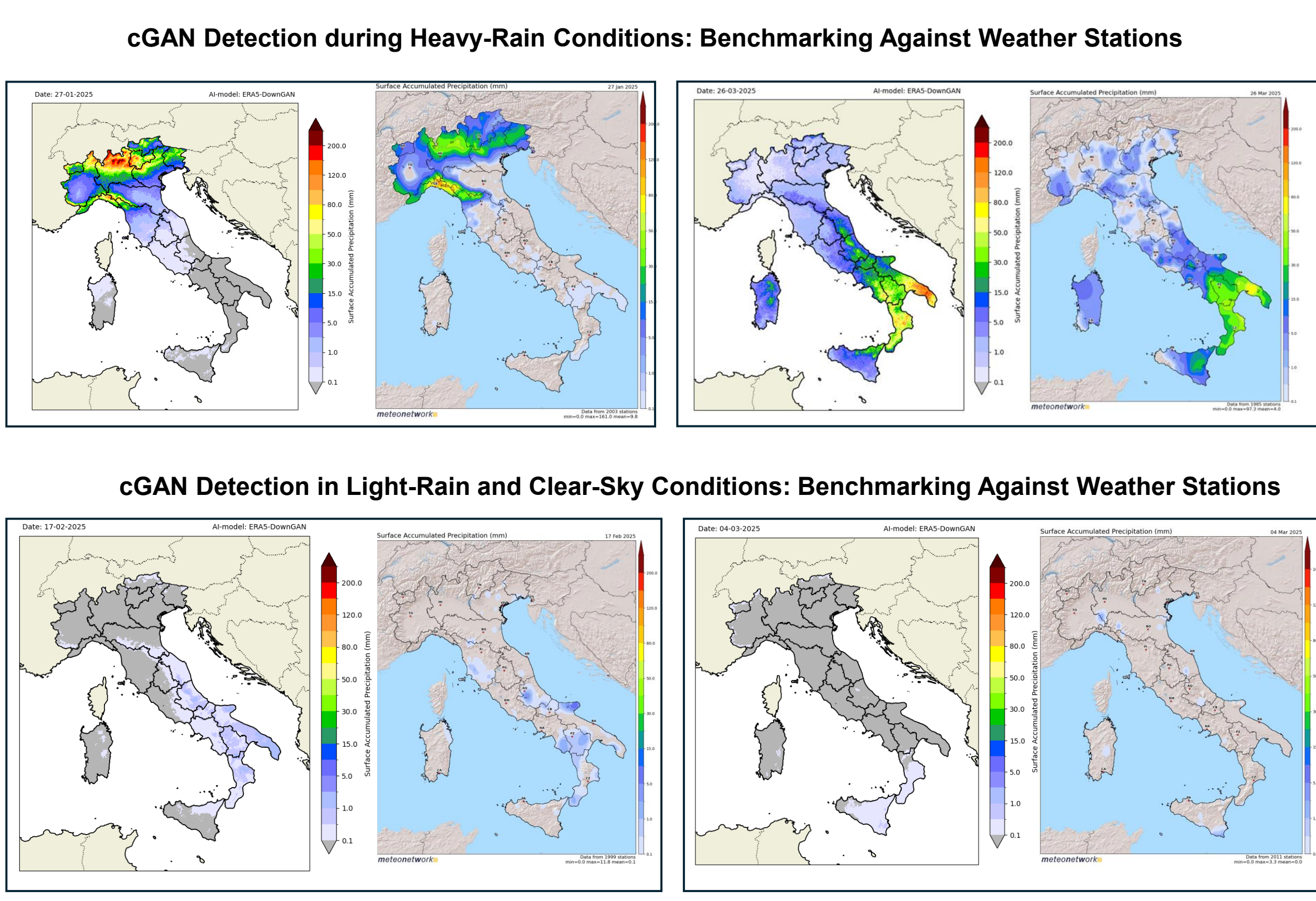
Training Dataset

- High resolution (HR)** : CONUS404 dataset (Rasmussen et al., 2023), at 4 km of horizontal resolution simulations of the WRF model.
- Low resolution (LR)** : ERA5 reanalysis data at a horizontal resolution of 0.25 degrees.

Fast AI Reanalysis: FAIR

FAIR is an innovative service developed by the CMCC Foundation to meet the growing need for timely, high-resolution climate data. It provides a high-resolution climate reanalysis for Italy (about 2.2 km), updated almost in real-time. It is based on downscaling data from the ERA5 reanalysis of the Copernicus program, one of the most widely used datasets in Europe for climate studies. FAIR offers downscaling of ERA5 data, reducing the update delay to just 6 days, compared to traditional methods that often take several months.

FAIR integrates dynamical downscaling dataset based on the **COSMO-CLM model** (VHR-REA_IT, [Raffa et al., 2021](#)), in an advanced deep learning technique, particularly a Conditional Generative Adversarial Network (cGAN), presented in [Manco et al. 2025](#).

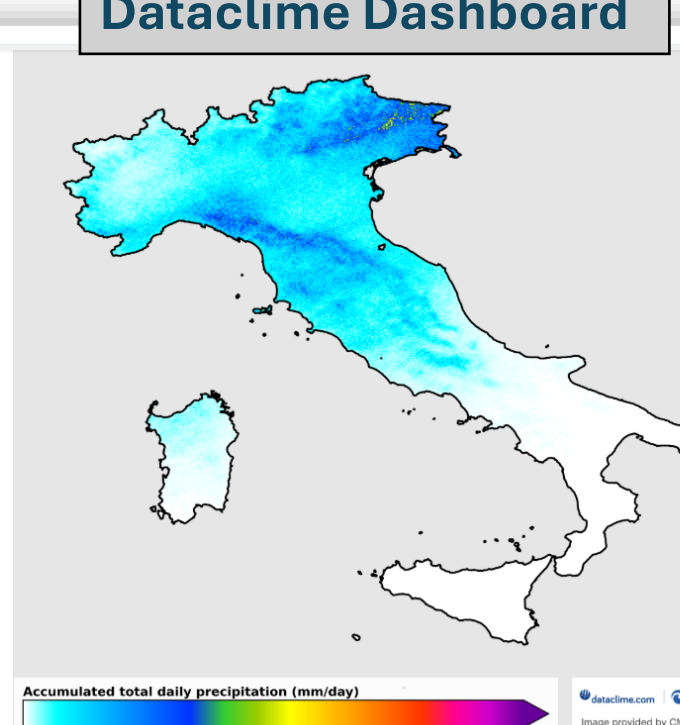


The service enables users to access high-resolution climate data (~2.2 km) with a **delay of only 6 days**, making it a valuable tool for rapidly **analyzing extreme events** and **enhancing responsiveness in adaptation and mitigation strategies**. All images are available for free download.

Next Steps

- Evaluation of the impact of using cross-validation windows and different low resolution predictors.
- Porting the model onto GPU for improved computational performance, also to test downscaling over large domains.
- Extending the downscaling approach to additional atmospheric variables, such as wind speed and pressure.
- Developing a semi-operational ERA5 downscaling framework.

Dataclimate Dashboard



References



Paper published